

Homework Specification

AI Technologies

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Resume checker hiring assist

Zoltán Szatmáry - (AIO6BZ)

szatmary.zoltan@edu.bme.hu

Introduction

The goal of the project is to develop an agentic LLM pipeline for automated CV screening with basic background verification. The system analyzes CVs against job descriptions (both provided as input from the user), flags mismatches, and produces structured scores with explanations. Data is sourced from public datasets and augmented synthetically; implementation combines multiple AI and NLP technologies.

Main Objectives

- *Data sources*
 - Public datasets from Kaggle and Hugging Face
 - Synthetic CVs for edge cases and adversarial testing
- *Design decisions*
 - Agent-based modular pipeline
 - Lightweight + heavy LLM separation
 - RAG for verification
 - Structured scoring to limit LLM variability
 - Basic bias mitigation via anonymization
- *Pipeline*
 1. Input (CV + job description)
 2. Preprocessing (extraction, cleaning, anonymization)
 3. Security agent (injection detection)
 4. Extraction (skills, experience, education)
 5. Verification (RAG-based checks)
 6. Evaluation (matching + scoring)
 7. Explanation (justified output)
- *Other objectives*
 - Reproducibility (fixed prompts/templates)
 - Modularity
 - Basic performance evaluation (accuracy, robustness)

Used Technologies

- **LangChain / LangGraph** – pipeline orchestration
- **Ollama** – local model execution
 - Lightweight model: security filtering
 - Larger model: reasoning and evaluation
- **FAISS** – retrieval for verification
- Other packages/libraries for web search and scraping, PDF text extraction, chunking, etc.